DESIGN AN INTERACTIVE AR/VR APPLICATION

Assignment Solution – AR/VR Development

Application Title: Virtual Science Lab – Immersive Learning for Students

Application Area: Education

Technology: Virtual Reality (VR)

Prepared as a basic design and prototype proposal

1. Introduction

Virtual Science Lab is an interactive Virtual Reality (VR) application that allows students to perform basic science experiments in a simulated laboratory. Students can explore laboratory equipment, follow experiment instructions, perform virtual actions, and receive immediate feedback without needing a physical laboratory for every practice session.

2. Problem Statement

Some schools and colleges have limited laboratory equipment, space, time, or budgets. Students may also be unable to repeat an experiment because materials are expensive or safety rules restrict repeated practice. This can make it difficult to understand practical concepts by reading textbooks alone. The proposed VR application provides a safe, repeatable, and interactive environment for learning laboratory procedures.

3. AR or VR Selection

Selected technology: Virtual Reality (VR). VR places the user inside a computer-generated 3D environment. In this application, the student enters a virtual science laboratory using a VR headset and interacts with simulated equipment. VR is suitable because the goal is to create a complete laboratory environment rather than overlay digital objects on the real world.

4. Objectives

• Help students understand science experiments through hands-on virtual practice.

• Allow learners to repeat activities without consuming real materials.

• Provide step-by-step instructions and instant feedback.

• Improve confidence before students perform experiments in a physical laboratory.

• Track experiment completion and quiz performance.

5. Main Features

Feature	Description
Virtual laboratory	A 3D laboratory containing benches, beakers, test tubes, measuring tools, and safety equipment.
Experiment selection	Students can choose an experiment such as measuring liquid volume or observing a simple chemical reaction.
Interactive objects	Users can point at, select, pick up, move, and place virtual tools.
Guided instructions	On-screen or audio instructions explain each step in the experiment.
Safety reminders	The application highlights protective equipment and warns users when a step is incorrect.
Instant feedback	The system explains mistakes and suggests the correct next step.
Progress dashboard	Completed experiments, scores, and practice history are displayed.
Quiz	A short quiz checks whether the student understood the experiment.

6. Required Hardware

• **VR headset:** A standalone headset such as Meta Quest, or a compatible PC VR headset.

• **Controllers:** Hand controllers for selecting and manipulating virtual objects.

• **Computer:** A VR-capable computer if the chosen headset requires a PC.

• **Optional equipment:** Headphones for spoken instructions and a safe, clear play area.

7. Required Software

• **Unity:** Game engine used to create the 3D laboratory and application logic.

• **XR Interaction Toolkit:** Supports VR interaction such as selecting, grabbing, and moving objects.

• **Blender (optional):** Used to create or edit 3D laboratory equipment models.

• **C#:** Used to program experiment steps, feedback, scoring, and progress tracking.

• **VR device runtime/SDK:** The software required by the selected headset.

8. Basic Design / Prototype

The prototype can contain four main screens or scenes:

• **Screen 1 – Welcome:** Application title, Start button, Instructions button, and Exit button.

• **Screen 2 – Experiment Menu:** Cards for available experiments, difficulty level, and estimated time.

• **Screen 3 – Virtual Lab:** 3D lab, interactive equipment, step instructions, safety reminders, and a progress indicator.

• **Screen 4 – Results:** Completion status, score, mistakes, and a button to retry or return to the menu.

Example prototype layout:

VIRTUAL SCIENCE LAB
Experiment: Measuring Liquid Volume
[3D LAB AREA: bench | measuring cylinder | beaker | water container]
Instruction: Select the measuring cylinder and pour water carefully.
[Hint] [Restart] [Next Step]

This is a screen-layout prototype, not a finished VR program. In a working prototype, the 3D lab would be built in Unity and the buttons and objects would be connected to interaction scripts.

9. How Users Interact with the Application

1. The student wears the VR headset and launches Virtual Science Lab.

2. The student selects an experiment from the menu using a controller.

3. The application shows the required equipment and safety instructions.

4. The student points at an object and presses the controller trigger to select it.

5. The student uses the grab control to pick up and move equipment, following the instructions.

6. The application checks the action. If it is incorrect, a hint appears; if it is correct, the next step is unlocked.

7. After finishing, the student completes a short quiz and views the result screen.

10. Example User Scenario

A student chooses the “Measuring Liquid Volume” experiment. The application asks the student to select a measuring cylinder, place it on the bench, and pour water from a virtual container. If the student selects the wrong tool, the application displays a hint. Once the steps are completed, the student answers a short quiz and receives a score with feedback.

11. Benefits

• Provides a safe space to practise procedures.

• Allows experiments to be repeated without using real consumables.

• Makes abstract or procedural concepts more engaging.

• Offers consistent instructions and feedback.

• Can help prepare students for practical laboratory sessions.

12. Limitations and Safety

A VR application cannot fully replace real laboratory experience. The simulation must be checked for scientific accuracy, and students still need real-world instruction for handling actual equipment. Some users may experience discomfort in VR, so sessions should be short, users should have a clear play area, and the application should provide an easy exit option. A headset and compatible hardware may also be costly.

13. Conclusion

Virtual Science Lab demonstrates how VR can be used in education to create an immersive and interactive learning experience. Students can explore a virtual laboratory, practise experiment steps, receive feedback, and review their progress. The proposed prototype covers the application area, problem statement, technology choice, main features, hardware and software, interface design, and user interaction.

14. Expected Outcome

By designing this application, students will understand the basic concept of Virtual Reality and how immersive technology can address practical learning challenges. They will also learn how to plan a simple VR interface, identify required development tools, and describe interactions between a user and a virtual environment.